CINNAMONSRI: A SMART IOT-INTEGRATED SYSTEM FOR ENHANCING CEYLON CINNAMON PRODUCTION AND EXPORT EFFICIENCY

25-26J-319

Project Proposal Report

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

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Abstract

Sri Lanka's cinnamon industry plays a vital role in the country's exports, but crop diseases such as leaf spot continue to threaten both yield and quality. Farmers often rely on manual inspections to identify diseases, a process that is not only slow and error-prone but also usually happens too late, once visible damage has already reduced productivity. To address this challenge, we propose a CNN-powered disease detection system designed specifically for cinnamon cultivation.

Through a simple mobile app, farmers can capture or upload images of cinnamon leaves or bark. These images are processed by a cloud-hosted CNN model, which identifies diseases such as leaf spot or blight and provides immediate results. Importantly, the system is also capable of detecting early-stage infections in nursery plants, allowing farmers to prevent the spread of disease before seedlings are transferred to larger fields. Recommendations are delivered in multiple formats—on-screen text, Sinhala/Tamil voice alerts, and expert-based treatment suggestions—making the system accessible to smallholder farmers across regions. Additionally, all detected cases are stored in a central database for monitoring trends and supporting long-term agricultural insights.

By enabling early intervention, from nursery to field level, especially during the critical rainy seasons when leaf spot outbreaks are most common, this AI-driven solution empowers farmers with real-time decision support to safeguard Ceylon cinnamon production

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1. INTRODUCTION

Sri Lanka, renowned as the "Spice Island," is the world's leading producer of Ceylon cinnamon (*Cinnamomum verum*) [1]. While the crop contributes substantially to export earnings, farmers face recurring threats from leaf spot and other fungal infections that flourish in humid conditions [2]. Manual detection depends heavily on farmer knowledge and visual inspection, often resulting in delayed diagnosis [3]. Studies highlight that cinnamon diseases can reduce yields by 20–30% annually if not addressed promptly [4].

This research proposes a smart, AI-powered disease detection module that integrates with the CinnamonSri platform, enabling real-time diagnosis via mobile devices. Using computer vision and CNN deep learning, the module addresses current gaps in scalability, accuracy, and accessibility of disease management for cinnamon farmers.

1.1 Background and Literature Survey

1.1.1 Background

Ceylon cinnamon, known worldwide for its unique aroma and flavor, is one of Sri Lanka's most valuable agricultural exports. The industry provides a livelihood for thousands of smallholder farmers and contributes significantly to the country's economy. However, cinnamon production faces many challenges beyond the post-harvest stage. Among them, plant diseases remain one of the most serious threats to both yield and quality.

One of the most common and damaging diseases is leaf spot, which spreads rapidly during the rainy season. Symptoms usually begin as small, dark spots on leaves that gradually enlarge, reducing the plant's ability to photosynthesize. If left untreated, severe infections can cause leaf drop, weaken the plant, and lower the overall quality of harvested cinnamon. Farmers typically rely on manual observation to identify the disease, but early symptoms are often overlooked until the infection has already spread. This delay leads to significant losses, especially for smallholders who lack access to expert advice or modern tools.

In recent years, technologies such as image processing and machine learning have shown promise in helping farmers identify plant diseases more accurately. With the growing availability of smartphones and affordable cameras, these technologies can now be made accessible directly to farmers. A mobile-based solution that detects cinnamon leaf spot disease quickly and reliably could reduce losses, support better disease management, and improve the consistency of cinnamon production in Sri Lanka.

1.1.2 Literature Survey

Researchers have explored the use of image processing and machine learning to detect diseases in cinnamon plants, but most existing studies are broad in scope and do not provide a specialized or farmer-centered solution for leaf spot disease. A closer look at previous work highlights both their contributions and their shortcomings.

- AgroX: Uplift Ceylon Cinnamon Industry

This project introduced modern technology to improve cinnamon production, including aspects of disease detection. However, its focus was on building a generalized platform for the industry rather than designing a system that tackles one specific disease. As a result, the disease detection component was not optimized for accuracy in real-world conditions.

- Cinnamon Care

This research attempted to digitize disease detection and provide treatment recommendations using image processing and machine learning. While it marked progress, its multi-disease approach reduced the ability to specialize and refine models for one high-impact disease such as leaf spot.

- The Cinnamon Therapist

This dissertation focused on stem and branch diseases rather than leaf-related conditions. While valuable, it overlooked leaf spot disease, which is more widespread among cinnamon plants and particularly harmful during rainy seasons.

- Systematic Literature Review (Dilhari & Sujah)

This work summarized existing methods and technologies used in cinnamon disease prediction. Although useful for understanding the research landscape, it was theoretical in nature and did not produce an applied system that farmers could use in practice.

From these studies, it is clear that while image-based disease detection has been studied in cinnamon, there is still a lack of disease-specific, stage-wise, and farmer-oriented tools. None of the above works combine a lightweight AI model, localized notifications, and offline support in a way that smallholder farmers can realistically adopt

1.2 Research Gap

The examination of current studies indicates that while there has been a focus on detecting cinnamon diseases, many of the proposed solutions are either overly generalized, excessively complicated, or not reasonable for smallholder farmers. A deeper analysis reveals several important gaps:

•Lack of disease-specific systems

Existing research often targets multiple cinnamon diseases at once. While this provides wider coverage, it reduces the accuracy and optimization possible for one specific disease. Leaf spot, which is the most common and damaging, has not been given focused attention.

•No severity-level classification

Most studies detect whether a plant is healthy or diseased, but they do not identify stages of infection (early, moderate, severe). This information is crucial for deciding whether preventive action or stronger treatment is required.

•Limited farmer accessibility

Current approaches are often research prototypes without strong consideration of how rural farmers will use them. Issues such as local language notifications, offline usability, and simple mobile interfaces are rarely addressed.

•High cost and technical barriers

Many systems depend on advanced hardware or cloud-based processing, which can be expensive or difficult for smallholder farmers to maintain. There is a lack of low-cost, lightweight solutions that can run on affordable mobile devices.

•Lack of contextual awareness

Cinnamon leaf spot is seasonal, with outbreaks becoming severe during the rainy months. Existing works do not include context-aware notifications to warn farmers ahead of high-risk periods.

Together, these gaps leave farmers without a practical, farmer-centered tool to detect and manage leaf spot disease effectively.

1.2.1 Research Gap and Novelty Analysis

To highlight the novelty of the proposed Leaf Spot Detection Module, a comparative analysis was carried out against six existing works. These include both cinnamon-focused research and general plant disease detection studies. The analysis compares features such as disease-specific focus, stage-wise detection, cost-effectiveness, offline usability, and localized notifications.

Feature	Agro X	Cinnamon Care	Cinnamon Therapist	Lit Review	Bandara	Proposed System

Focus on cinnamon	Yes	Yes	Yes (stem/branch)	Yes	Yes	Yes
Focus on leaf spot	Partial	Partial	No	No	Partial	Yes
Stage-wise Detection	No	No	No	No	Partial	Yes
Treatment Recommendat-ions	Yes	Yes	Yes	No	Yes	Yes(stage-based)
Low cost for smallholders	No	No	No	No	No	Yes
Localized notifications	No	No	No	No	No	Yes
Practical Farmer-Centric Design	Partial	Partial	Partial	No	Partial	Yes

Table 1: Comparative Analysis Highlighting the Novelty of the Proposed System

Explanation of Research Gap and Novelty

- Research Gap: Existing cinnamon-specific works (AgroX, Cinnamon Care, Cinnamon Therapist, Literature Review, Bandara) either cover multiple diseases or focus on different plant parts, without specializing in leaf spot. General plant disease studies (Sharma, 2022) show technical feasibility but lack spice-specific datasets, local adaptation, or usability for smallholders. None of the works address severity-level classification, offline deployment, or localized notifications.

- Novelty of Proposed System: This project is the first to specialize exclusively in cinnamon leaf spot disease with a CNN-based approach that not only detects infection but also classifies early, moderate, and severe stages. It provides stage-based treatment guidance, localized notifications in Sinhala/Tamil, and offline-ready mobile deployment using TensorFlow Lite. These features directly address the gaps, making the solution both technically robust and practically usable by cinnamon farmers in Sri Lanka

1.3 Research Problem

Cinnamon farmers in Sri Lanka still rely on **manual observation** to identify diseases in their crops. In the case of **leaf spot disease**, the most common cinnamon leaf infection, symptoms often go unnoticed until the disease reaches a severe stage. This leads to lower leaf quality, premature leaf drop, and reduced yield. The absence of **early and stage-wise detection methods** makes it difficult for farmers to take timely preventive action.

Although previous research has applied image processing and machine learning to cinnamon disease detection, these systems are either generalized across multiple diseases or require high technical expertise and resources. They do not offer **specialized and farmer-friendly solutions** that smallholders can realistically use. Moreover, none of the existing systems combine **disease-specific models, severity classification, localized notifications, and practical treatment recommendations**

2. OBJECTIVES

2.1 Main Objective

Provide farmers with cutting-edge tools for better quality, price, and industry decision-making by building a comprehensive AI/ML-based platform to increase Ceylon cinnamon production and export efficiency in Sri Lanka.

2.2. Specific Objectives

The following sub-objectives guide the development of this research:

1. **Collect and preprocess image datasets** of healthy and diseased cinnamon leaves with emphasis on leaf spot during different growth stages.
2. **Train and validate a Convolutional Neural Network (CNN)** model to classify leaves as healthy, early-infected, moderately-infected, or severely-infected.
3. **Integrate the trained model into a mobile application** that allows farmers to capture or upload images of cinnamon leaves.
4. **Provide localized notification alerts** (Sinhala/Tamil) with stage-specific treatment or prevention advice based on model output.

Evaluate the system's accuracy and usability through testing with real-world farmer data and feedback.

3. METHODOLOGY

3.1 System Overview

The CinnamonSri system is a comprehensive, integrated platform that uses the Internet of Things (IoT), machine learning (ML), artificial intelligence (AI), and user-friendly mobile applications to transform Ceylon cinnamon production in Sri Lanka. This solution's four interrelated components help smallholder farmers increase productivity, quality, and profitability while addressing major issues in the cinnamon sector, including uneven drying, manual grading, price volatility, and disease control.

1. Smart Drying Automation (IoT): This part uses DHT22 sensors to track the humidity and temperature in drying chambers in real time. An ESP8266 microcontroller regulates fans and heaters via relays to maintain optimal conditions. Data is sent to Firebase for storage and display via a mobile application that offers voice notifications in Sinhala. This minimizes deterioration from over- or under-drying and guarantees consistent export-grade quality and remote monitoring.

2. AI-Powered Quality & Price Prediction: Farmers use a mobile or online app to enter characteristics like as temperature, moisture content, bark thickness, and pre- and post-drying weights. ML models predict quality ratings (Premium, Grade A, and Grade B) and anticipated costs (LKR/kg). Examples of these models are Random Forest for classification and XGBoost for regression. This reduces dependency on middlemen and gives farmers data-driven insights to negotiate better prices.

3. A Guide to AI Cinnamon Harvesting: This AI-powered module uses trained models on cinnamon growth stage datasets to assess uploaded plant photos. By using regional planting calendars to determine maturity and health condition, it sends out time-based notifications. Alerts guide farmers on planting, care, and harvesting, optimizing yield and reducing losses from premature harvesting.

4. Disease Detection Module: Using the app, farmers can take or submit pictures of leaves or bark. Leaf spot is classified by a Convolutional Neural Network (CNN) model that was trained on datasets related to cinnamon sickness. Treatment recommendations are included in the results, along with voice notifications in Tamil and Sinhala for prompt action. Early intervention is made possible by the logging of data to a central dashboard for long-term monitoring, particularly during wet seasons.

These elements work together seamlessly through a cloud backend (like Firebase) and centralized mobile app, enabling data exchange and multilingual user interfaces. CinnamonSri fills technical gaps by integrating automation, predictive analytics, and localized alerts. This promotes sustainable practices and boosts financial returns for Sri Lanka's cinnamon industry. Figure 1 illustrates the CinnamonSri platform and its four interrelated components clearly.

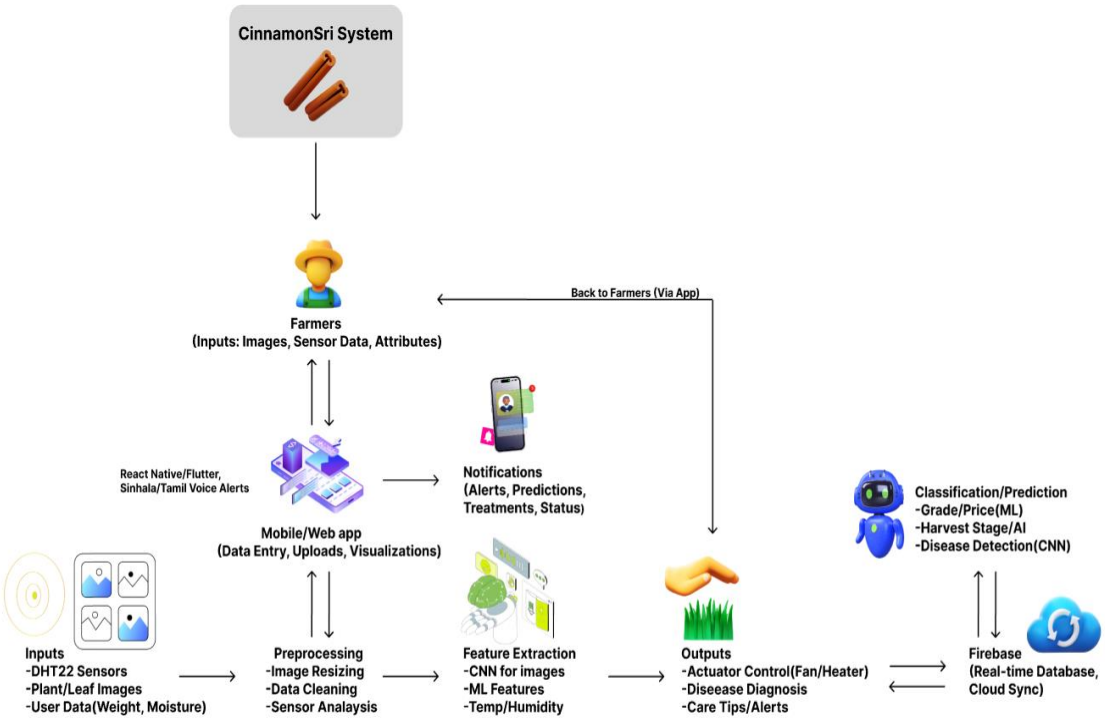

Figure 1: System Overview Architecture

3.2 Individual Component Overview

The Leaf Spot Disease Detection Module is developed as a specialized component of the CinnamonSri system. Its purpose is to help cinnamon farmers identify the presence and severity of leaf spot disease through a mobile-based application powered by artificial intelligence.

Farmers capture or upload images of cinnamon leaves using their smartphones. These images are processed by a Convolutional Neural Network (CNN) model trained on datasets of healthy and infected leaves. The model classifies the input image into one of four categories:

- Healthy
- Early-stage infection
- Moderate infection
- Serious infection

Based on the identified stage, the system generates localized notification alerts (Sinhala or Tamil) that provide farmers with stage-specific recommendations. For example, early-stage infections may prompt preventive actions such as pruning, while serious cases may recommend fungicide application or isolation of the plant.

The module directly addresses the challenges of manual observation, where farmers often misidentify or delay the recognition of early symptoms. By introducing a simple mobile interface combined with advanced CNN-based image analysis, the system ensures that even farmers with limited technical skills can use it effectively. This approach enables quicker intervention, reduces the spread of disease, and ultimately supports more sustainable cinnamon farming practices in Sri Lanka

3.3 Requirement Gathering

3.4 Past Research Analysis

3.5 Feasibility Study

3.5.1 Technical Feasibility

3.5.1.1 Knowledge on Technologies

3.5.1.2 Knowledge on Tools

3.5.1.3 Data collection Knowledge

3.5.2 Schedule Feasibility

3.5.3 Economic Feasibility

3.6 System Analysis

3.6.1 Software Solution Approach

.

Figure 3: General Flow of Cinnamon Quality and Price Prediction

3.6.2 Tools & Technology

- Programming Languages: Python
- Frameworks: scikit-learn, XGBoost, TensorFlow (for potential deep learning enhancements), Flask (web app), Flutter (mobile app)
- Database Systems: Firebase (for storing farmer input data and predictions)
- Web Development Frameworks
  - React JS: Front-end development for the web interface
  - Tailwind CSS: UI component design and development
- UI Design Implementation: Figma

- Diagramming Too: Draw.io
- Integrated Development Environment (IDE): VS Code
- Version Control System (VCS): Git (GitHub)
- Tools: Microsoft Teams

- Testing Tools
  - Pytest: Unit testing for Python
  - Postman: API testing
- Deployment Tools
  - Docker: Containerization
  - Kubernetes: Orchestration
  - Jenkin: Automation
  - Ansible: Configuration
- Code Quality Assurance Too: SonarQube
- Project Management: MS Project

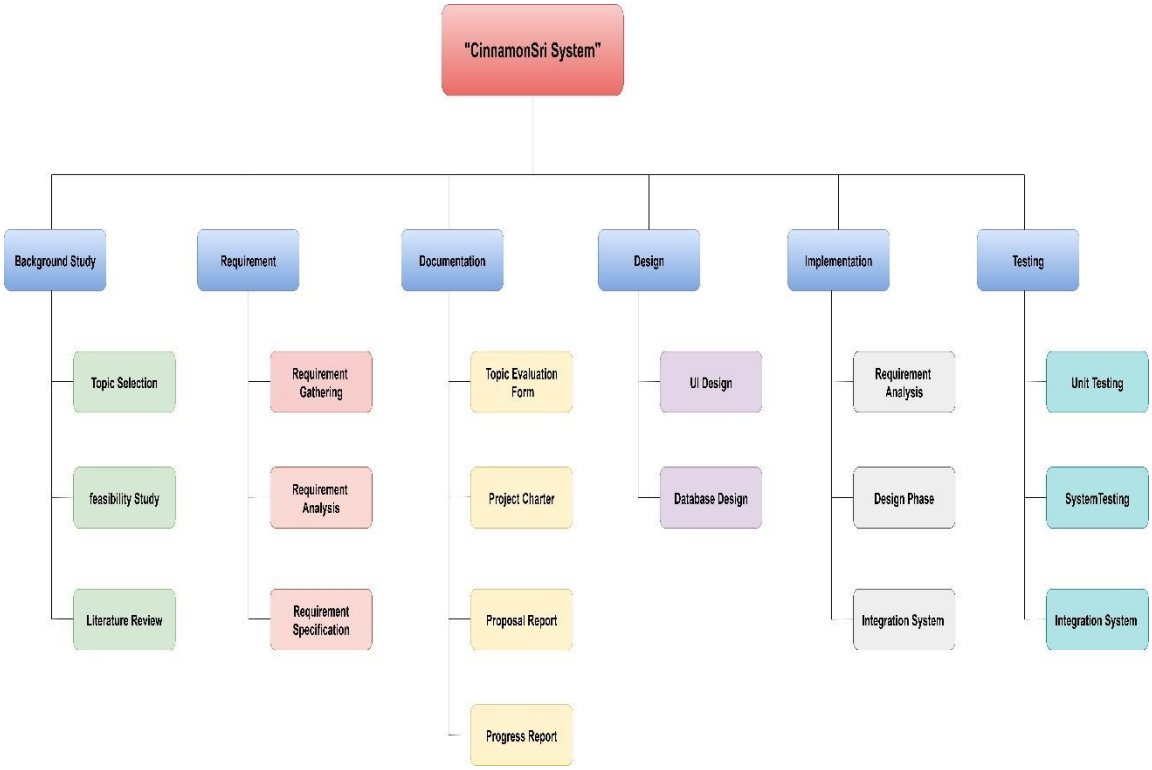

Figure 4: Work breakdown Structure

3.7 Project Requirements

3.7.1 Functional Requirements

3.7.2 Non-Functional Requirements

3.8 Testing

Table 2: Test Cases

3.9 Timeline

The proposed timeline for the project is as follows.

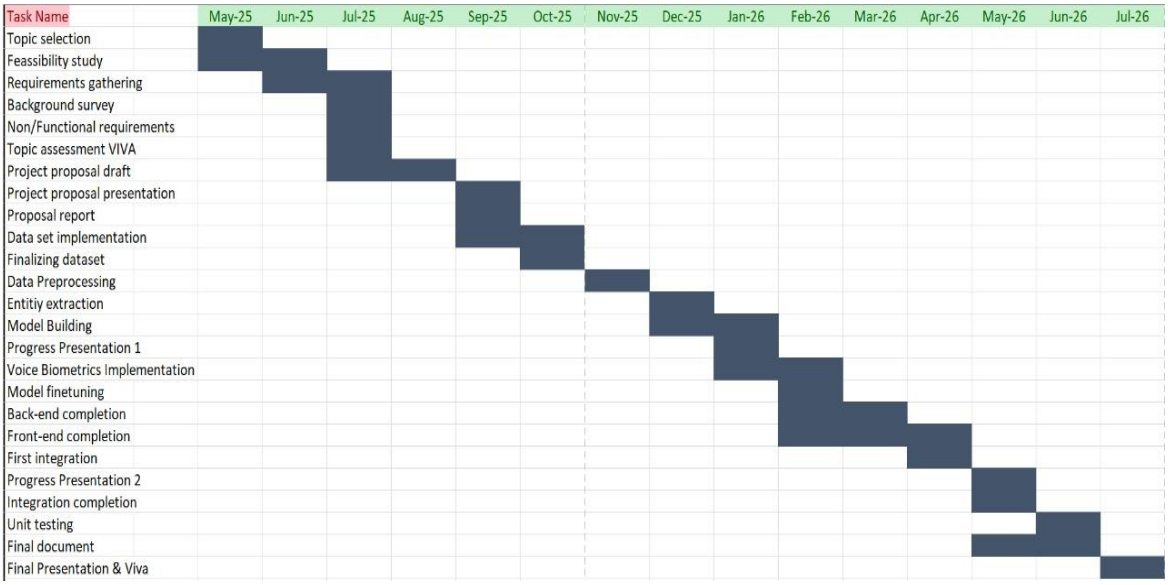

Figure 5: Gantt Chart for Project Schedule Management

4.0 BUDGET

5.0 SUMMARY

References